

main.py

Visualize

Run

```
import numpy as np
temp = np.array([25, -3, 28, -1, 30, 27, -5])
temp[temp < 0] = 0
print("Updated values:", temp)
```

Output

Updated values: [25 0 28 0 30 27 0]

=== Code Execution Successful ===

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```
1 import numpy as np
2
3 rainfall = np.array([
4     120, 85, 150, 200, 175, 90,
5     110, 160, 210, 180, 95, 130
6 ])
7 total = np.sum(rainfall)
8 average = np.mean(rainfall)
9 months = np.where(rainfall > average)[0] + 1
10 print("Total Rainfall:", total)
11 print("Average Rainfall:", round(average, 2))
12 print("Months having rainfall above average:")
13 print(months)
```

```
Total Rainfall: 1705
Average Rainfall: 142.08
Months having rainfall above average:
[ 3  4  5  8  9 10]
```

```
=== Code Execution Successful ===
```

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```
1 import numpy as np
2
3 matrix = np.array([
4     [1, 2, 3],
5     [4, 5, 6],
6     [7, 8, 9]
7 ])
8 row_sums = np.sum(matrix, axis=1)
9 column_sums = np.sum(matrix, axis=0)
10 print("Row sums:", row_sums)
11 print("Column sums:", column_sums)
```

Row sums: [6 15 24]
Column sums: [12 15 18]

=== Code Execution Successful ===

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```
1 import numpy as np
2
3 test1 = np.array([65, 80, 72, 90, 55])
4 test2 = np.array([70, 75, 78, 85, 60])
5
6 students = np.where(test2 > test1)[0] + 1
7
8 print("Students improved:", students)
```

Output

Students improved: [1 3 5]

=== Code Execution Successful ===

main.py

Visualize

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Output

```
import numpy as np

marks = np.array([78, 65, 92, 55, 84])

print("Average:", np.mean(marks))
print("Highest:", np.max(marks))
print("Lowest:", np.min(marks))
```

Average: 74.8
Highest: 92
Lowest: 55

=== Code Execution Successful ===